

Use of Artificial Intelligence–Based Detection of Diabetic Retinopathy in the US

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Vision loss from diabetic retinopathy (DR) is largely preventable, but less than two-thirds of patients with type 1 or type 2 diabetes undergo an annual eye examination.¹ FDA-approved systems like LumineticsCore (formerly IDx-DR, Digital Diagnostics) and EyeArt (Eyenuk) analyze retinal fundus images for more than mild DR with fairly high sensitivity (87.2% and 96.0%, respectively) and specificity (90.7% and 88.0%, respectively).^{2,3} Use of these systems helps increase detection in the primary care setting among patients with diabetes, while optimizing ophthalmic examinations for those with vision-threatening DR. We tracked usage of *Current Procedural Terminology* code 92229, the artificial intelligence (AI)–based screening reimbursement code instituted in January 2021, to evaluate national trends of AI-based DR detection.

Methods

A retrospective cohort study was performed using the TriNetX federated database, which encompasses more than 107 million patients across 62 health care organizations in the US. Records for patients with diabetes were examined from January 2019 to December 2023, with data analysis completed in May 2024. Use rates (measured per 100 000 patients with diabetes) of code 92229 were compared with traditional codes for remote eye imaging (92227 and 92228) and imaging modalities from secondary referrals, including fundus photography (92250) and optical coherence tomography (OCT, 92134), with no adjustment to *P* values for multiple analyses.⁴ This study was exempt from approval by the Stanford University institutional review board, as TriNetX deidentifies all patient information, and followed [STROBE](#) reporting guidelines. Statistical analysis was performed in Python version 3.8 (Python Software Foundation).



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Of 4 959 890 patients with diabetes in the TriNetX system from January 2019 to December 2023, 209 673 unique patients (4.2%) received at least 1 of the targeted codes. Mean (SD) patient age was 64 (16) years, and 2 380 747 patients (48.0%) were female. Of patients with at least 1 targeted code, OCT imaging (92134) was used in 168 382 patients (80.3%), fundus photography (92250) in 73 363 patients (35.0%), and traditional remote imaging (92227 and 92228) in 2135 patients (1.0%) and 5232 patients (2.5%), respectively. Since 2021, AI imaging (92229) was used in 3440 of 154 136 cases (2.2%), meaning that only 0.09% of all patients with diabetes received this new modality for DR detection. In 2021, AI imaging (92229) was used 58.0 times per 100 000 patients, which marginally increased by 1.0% to 58.6 times per 100,000 patients by 2023. Although use of traditional remote imaging increased by 185.4% between 2021 and 2023, AI imaging had a higher referral rate (7.74%) to OCT imaging than traditional remote imaging (5.53%). Use of all remote imaging modalities increased by 90.16% from 2021 to 2023 (95% CI, 69.60%-110.72%; $P < .001$; [Table 1](#)). More than 80% of those who received AI imaging were from the South, a region that made up only 40% of other imaging modalities, and almost half of the patients who received AI imaging were Black, compared with approximately a quarter in other imaging modalities ([Table 2](#)).

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Table 1. Use per 100 000 Patients With Diabetes (Types 1 and 2)

 [Use per 100 000 Patients With Diabetes \(Types 1 and 2\)](#)

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Table 2. Baseline Characteristics of Patients Who Underwent Each Imaging Modality From January 1, 2019, to December 31, 2023

 [Baseline Characteristics of Patients Who Underwent Each Imaging Modality From January 1, 2019, to December 31, 2023](#)

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Discussion

The FDA's approval of AI-based systems may be a step forward in DR detection, although adoption is nascent and traditional remote monitoring methods for DR remain more prevalent. This cohort study, while not population based, found that only 4.2% of diabetic patients received ophthalmic imaging for DR over 5 years. Imaging rates may be artificially low due to eye care professionals not routinely performing ancillary ophthalmic imaging during general diabetic eye examinations. Despite AI-based imaging leading to more OCT referrals compared to traditional methods, barriers persist, such as cost, awareness, integration, and FDA approval of AI software for imaging devices.⁵ This potential un-

Programs like the Stanford Teleophthalmology Autonomous Testing and Universal Screening program highlight AI's potential in improving DR detection but also highlight the importance of streamlined workflows, close collaboration between primary care and ophthalmology, and patient-friendly appointment scheduling.⁶ These findings support further evaluation of imaging practices to develop targeted strategies for improving diabetic eye imaging rates and patient outcomes.

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